

Demonstration of Proposed Features

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	User Input Form	HTML/CSS interface allowing users to input the 4 core socio-economic metrics.	Implemented	Yes	Fully functional and styled on the live website.
2	Input Validation	Logic to ensure entered values are within logical bounds before processing.	Implemented	Yes	Prevents backend crashes from invalid data entries.
3	Data Normalization	Processes and scales the raw user inputs to match the ML training data format.	Implemented	Yes	Operates invisibly in the Flask backend using <code>scaler.pkl</code> .
4	HDI Prediction Engine	Executes the Scikit-learn Linear Regression model to calculate the final HDI score.	Implemented	Yes	Successfully predicts scores with high accuracy ($R^2 = 0.981$).
5	Results Dashboard	Dynamically displays the predicted HDI score and development tier category.	Implemented	Yes	Updates seamlessly via JavaScript without reloading the page.
6	REST API Endpoint	Flask routing (<code>/predict</code>) that receives JSON data and returns predictions.	Implemented	Yes	Verified via Postman and frontend integration.

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	8
Total Features Implemented	7
Total Features Demonstrated	7
Overall Implementation Rate (%)	87.5%